

B.Tech. Degree V Semester Supplementary Examination
April 2019

ME 15-1503 MACHINE TOOLS AND MACHINING SCIENCE
(2015 Scheme)

Time: 3 Hours

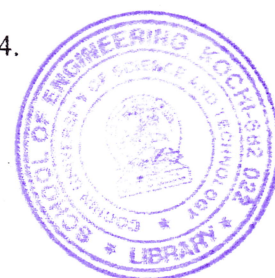
Maximum Marks: 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I.
 - (a) Describe tool holding devices in Capstan and Turret lathes.
 - (b) Briefly describe a Jig boring machine.
 - (c) How do you get the reciprocating motion in a slotting machine?
 - (d) What are the different methods of indexing in milling?
 - (e) The spindle end of a milling machine arbor has a taper of 7:24. Determine the setting of the compound rest.
 - (f) Describe Jigs and fixtures for drilling operations.
 - (g) Briefly describe various types of chips in machining.
 - (h) With example describe multipoint cutting tools.
 - (i) Write a note on variables affecting tool life.
 - (j) Briefly explain Lee and Shaffer's relationship.



PART B

$$(4 \times 10 = 40)$$

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|-------|-----|--|------|
| II. | (a) | Write the complete specification for purchasing a Lathe. | (5) |
| | (b) | Distinguish between Shaper and Slotter. | (5) |
| | | OR | |
| III. | | With sketch explain any one quick return mechanism used in a Shaping machine. | (10) |
| IV. | (a) | Write a note on cylindrical and surface grinding machines. | (5) |
| | (b) | Calculate all machining particulars for cutting a spur gear of 3 module and 54 teeth with proper index plate hole circle and sector. | (5) |
| | | OR | |
| V. | | Describe the principles of location and clamping in Jigs and Fixtures. | (10) |
| VI. | (a) | Briefly explain the mechanism of built-up-edge formation. | (5) |
| | (b) | In an orthogonal machining operation, the rake angle of the tool was 15° , shear angle 35° . Calculate the shear strain. | (5) |
| | | OR | |
| VII. | | Explain the thermal aspects of machining in detail with neat sketches. | (10) |
| VIII. | (a) | Write a note on tool wear and tool life. | (3) |
| | (b) | Explain the relationship between shear angle, angle of friction and cutting rake angle developed by Merchant. | (7) |
| | | OR | |
| IX. | (a) | Describe Flank wear and Crater wear. | (5) |
| | (b) | Briefly explain mechanism of tool wear. What do you understand by diffusion wear? | (5) |